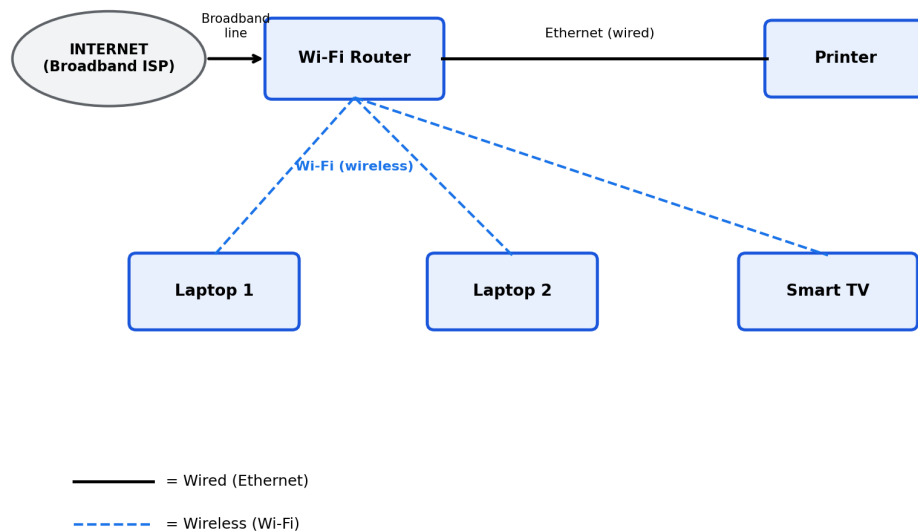


Assignment : Network Security & Management

Topic : Build a SOHO Network

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

SOHO (Small Office / Home Office) Network Setup



- For this setup I connected the Printer to the router by Ethernet cable (wired) since it stays in a fixed spot and wired gives a more reliable print connection, while both Laptops and the Smart TV connect over Wi-Fi (wireless) since they move around or don't have an easy cable run to the router.
2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.
Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.
- Broadband connection – a fibre connection such as Jio Fiber or Airtel Xstream Fiber (the ISP provides an ONT/modem box).
 - Wi-Fi Router – TP-Link Archer C6 (AC1200 dual-band router), easily available on Flipkart/Amazon and enough for a small home setup.
 - Ethernet cable – one CAT6 cable (~2–3 m) to wire the printer to the router.
 - Printer – HP DeskJet 2331 or similar, connected via the Ethernet/USB port.
 - Switch/extender – not required for just 5 devices since the router's built-in 4-port LAN switch is enough; if more wired devices were added later, a TP-Link TL-SG105 5-port switch could be plugged into a spare LAN port.
 - A power strip/surge protector for the router and modem.

Assignment : Network Security & Management

3. **Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.**
 1. Connect a laptop to the router (via Ethernet or its default Wi-Fi) and open a browser to the router's default gateway IP, e.g. 192.168.0.1 or 192.168.1.1.
 2. Log in using the default admin username/password printed on the router's label.
 3. Go to Wireless Settings and change the SSID from the default name to something unique, e.g. "MyHome_5G".
 4. Set the security mode to WPA2-PSK (AES) and choose a strong Wi-Fi password.
 5. Go to the Administration/System tab and change the default admin password to a new, strong password.
 6. Save the settings, reboot the router, and reconnect all devices using the new SSID and password.
4. **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

Device	MAC Address (example)	IP Address	Type
Wi-Fi Router	—	192.168.1.1	Gateway
Printer	A4:5E:60:B2:3C:11	192.168.1.10	Static
Smart TV	5C:E0:C5:7A:9F:22	192.168.1.11	Static
Laptop 1	88:E9:FE:12:3D:45	192.168.1.100	DHCP
Laptop 2	3C:A0:67:9B:1E:78	192.168.1.101	DHCP

5. **Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.**
 - Wrong or outdated Wi-Fi password saved on the laptop → forget the saved network on the laptop and reconnect, entering the correct WPA2 password fresh.
 - IP address conflict or the router's DHCP not assigning an address → run ipconfig and check for a 169.254.x.x (APIPA) address; if seen, release/renew the IP (ipconfig /release then /renew) or restart the router.
 - Router/ISP-side outage, or the laptop is out of Wi-Fi range → check whether other devices can connect; if none can, restart the modem/router; if only this laptop fails, move it closer to the router or check its Wi-Fi adapter/driver.

Assignment : Network Security & Management

Topic : Secure Networks with Firewalls, NAT, Port Filtering, and Other Methods

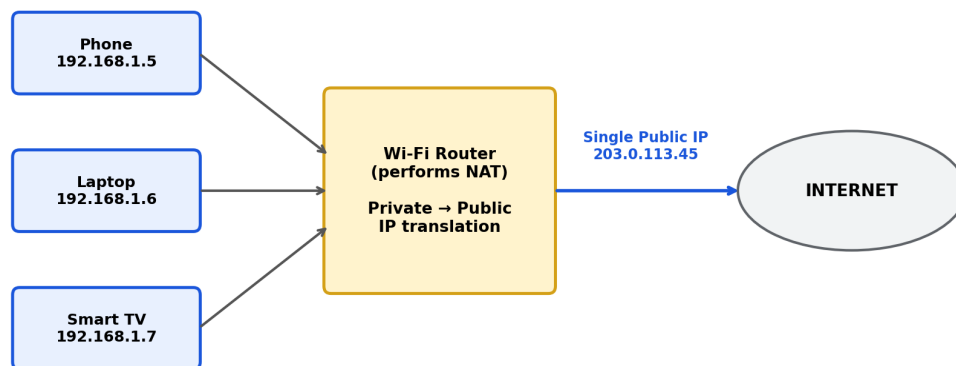
1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.
 1. Opened Windows Defender Firewall with Advanced Security (search “wf.msc” in the Start menu).
 2. Selected Inbound Rules → New Rule... in the right-hand panel.
 3. Chose Port as the rule type, then TCP and Specific local ports: 8080.
 4. Selected Block the connection, applied it to all profiles (Domain, Private, Public), and named the rule “Block Port 8080”.
- The rule now appears in the Inbound Rules list with these properties:

Property	Value
Rule name	Block Port 8080
Protocol	TCP
Local port	8080
Action	Block
Profile	Domain, Private, Public

2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.

NAT (Network Address Translation) - Home Wi-Fi Router



All 3 devices share ONE public IP (203.0.113.45) when talking to the internet.
The router keeps a NAT table mapping each private IP:port back to the right device.

Assignment : Network Security & Management

3. **Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**

- Logged into the router's admin panel → Firewall / Port Filtering section, and added a new outbound rule blocking TCP/UDP port 25 for all devices on the LAN.
- Blocking outbound port 25 stops any infected device on the network from being used as a spam relay that sends bulk email directly over raw SMTP. It doesn't affect normal email use, since modern email apps submit mail through the provider's authenticated servers on port 587/465, not port 25.

4. **Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.**
Hint: Focus on source/destination IPs and ports in the rule set.

- The exact rule set wasn't attached here, so I worked through this using a typical basic firewall rule set (swap in the trainer's actual rules if different):

Rule #	Rule
1	Allow outbound TCP any → any : 80, 443 (HTTP/HTTPS)
2	Allow outbound UDP any → any : 53 (DNS)
3	Block inbound/outbound TCP any → any : 22 (SSH) from untrusted sources
4	Deny all other traffic (default deny)

Packet	Matches	Verdict
1) HTTP, 192.168.1.5 → 172.217.166.78:80	Rule 1 (port 80)	ALLOWED
2) SSH, 10.0.0.10 → 34.201.23.1:22	Rule 3 (port 22 blocked)	BLOCKED
3) DNS, 192.168.1.7 → 8.8.8.8:53	Rule 2 (port 53)	ALLOWED

5. **Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.**

- A Web Application Firewall (WAF) in front of their servers to filter malicious HTTP requests (like SQL injection or XSS attempts) before they ever reach the application, e.g. AWS WAF or Cloudflare.
- Rate limiting / IP-based packet filtering that blocks or throttles IP addresses making an abnormally high number of requests, to protect against DDoS attacks and scraping bots.
- Restricting inbound access to internal admin/database ports so only specific internal IP ranges or VPN traffic can reach them, while only standard web ports (443) are exposed to the public internet.

Assignment : Network Security & Management

Topic : Authentication and Access Control

1. Create a simple login form for a mock Zomato admin panel that asks for username and password, then displays 'Access Granted' only if both fields match hardcoded admin credentials.

```
<!-- login.html -->
<form id="loginForm">
  <input id="username" placeholder="Username" />
  <input id="password" type="password" placeholder="Password" />
  <button type="submit">Login</button>
</form>
<p id="result"></p>

<script>
const ADMIN_USER = "admin";
const ADMIN_PASS = "Zomato@123";

document.getElementById("loginForm").addEventListener("submit", function (e) {
  e.preventDefault();
  const u = document.getElementById("username").value;
  const p = document.getElementById("password").value;
  document.getElementById("result").textContent =
    (u === ADMIN_USER && p === ADMIN_PASS) ? "Access Granted" : "Access Denied";
});
</script>
```

2. Write a Python function `check_access(user_role, resource)` that returns `True` if the `user_role` is allowed to access the resource, otherwise `False`. Test it with roles like 'customer', 'restaurant_owner', and resources like 'menu_edit', 'order_history', 'admin_dashboard'.
Hint: Use a dictionary to map roles to allowed resources.

```
ROLE_PERMISSIONS = {
    "customer": ["order_history"],
    "restaurant_owner": ["menu_edit", "order_history"],
    "admin": ["menu_edit", "order_history", "admin_dashboard"],
}

def check_access(user_role, resource):
    return resource in ROLE_PERMISSIONS.get(user_role, [])

# Test cases
print(check_access("customer", "order_history"))    # True
print(check_access("customer", "menu_edit"))        # False
print(check_access("restaurant_owner", "menu_edit")) # True
print(check_access("restaurant_owner", "admin_dashboard")) # False
print(check_access("admin", "admin_dashboard"))    # True
```

3. Simulate a two-factor authentication flow like Paytm: after a user enters the correct password, generate a 6-digit OTP and display it in the console. Ask the user to input the OTP and display 'Login Successful' only if it matches.

```
import random

STORED_PASSWORD = "MyPaytm@2024"

entered_password = input("Enter password: ")
if entered_password == STORED_PASSWORD:
```

Assignment : Network Security & Management

```
otp = random.randint(100000, 999999)
print(f"Your OTP is: {otp}") # in a real app this would be sent via SMS, not
printed
entered_otp = input("Enter the OTP: ")
if int(entered_otp) == otp:
    print("Login Successful")
else:
    print("Incorrect OTP")
else:
    print("Incorrect password")
```

4. Given an array of user objects with properties: username, isPremium, and lastLogin, write a function that returns only those users who are premium and have logged in within the last 7 days.

Constraint: Assume lastLogin is a date string in 'YYYY-MM-DD' format, and today is '2024-06-10'.

```
const TODAY = new Date("2024-06-10");

function getActivePremiumUsers(users) {
  return users.filter((user) => {
    if (!user.isPremium) return false;
    const lastLogin = new Date(user.lastLogin);
    const daysSince = (TODAY - lastLogin) / (1000 * 60 * 60 * 24);
    return daysSince <= 7;
  });
}

const users = [
  { username: "riya", isPremium: true, lastLogin: "2024-06-08" },
  { username: "aman", isPremium: true, lastLogin: "2024-05-20" },
  { username: "zoya", isPremium: false, lastLogin: "2024-06-09" },
];

console.log(getActivePremiumUsers(users));
// Output: [ { username: 'riya', isPremium: true, lastLogin: '2024-06-08' } ]
```

5. Use ChatGPT to generate a list of 5 common access control mistakes developers make in real-world apps like Instagram or Flipkart, and for each, write a one-line fix or prevention tip.

Mistake	Fix / Prevention Tip
Checking permissions only in the frontend (UI hides a button but the API still allows the action)	Always re-check the user's role/permissions on the backend for every request, not just in the UI.
Using predictable resource IDs (e.g. /user/1002/orders) without checking ownership	Verify the logged-in user actually owns or is allowed to access that specific resource ID before returning data.
Giving new users overly broad default roles/permissions	Follow the principle of least privilege — grant only the minimum access a role needs by default.
Forgetting to revoke access after a role change (e.g. ex-employee still has admin rights)	Automatically re-evaluate or expire permissions whenever a user's role changes or their account is deactivated.
Hardcoding admin credentials or leaving default passwords unchanged	Enforce strong, unique credentials and remove/rotate any default admin accounts before going live.

Assignment : Network Security & Management

Topic : Network Threats and Mitigation & Physical Security and Risk

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Threat	Real-World Example
Phishing	Fake SMS/emails pretending to be from Paytm, asking users to “verify” their account by entering their password/OTP on a fake login page.
DDoS	Attackers flooding a service like Instagram's servers with huge amounts of fake traffic to try to take it offline.
Man-in-the-Middle	An attacker on public Wi-Fi intercepting traffic between a user's phone and a website if the connection isn't properly encrypted (HTTPS).
Malware	A fake “WhatsApp Plus” APK downloaded outside the Play Store that secretly installs spyware/a keylogger on the phone.
Password Attacks	Attackers using leaked password lists from other breaches (credential stuffing) to try logging into a user's Zomato account with a reused password.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.
Constraint: The script should print the number of attempts it took to guess the correct PIN.

```
ACTUAL_PIN = "4821" # the "secret" PIN being brute-forced
```

```
attempts = 0
for i in range(10000):
    guess = f"{i:04d}" # 0000, 0001, ... 9999
    attempts += 1
    if guess == ACTUAL_PIN:
        print(f"PIN found: {guess}")
        print(f"Number of attempts: {attempts}")
        break
```

- This shows why 4-digit PINs are weak on their own — with no lockout after failed attempts, all 10,000 combinations can be tried almost instantly, which is why real systems add attempt limits, delays, or 2FA.

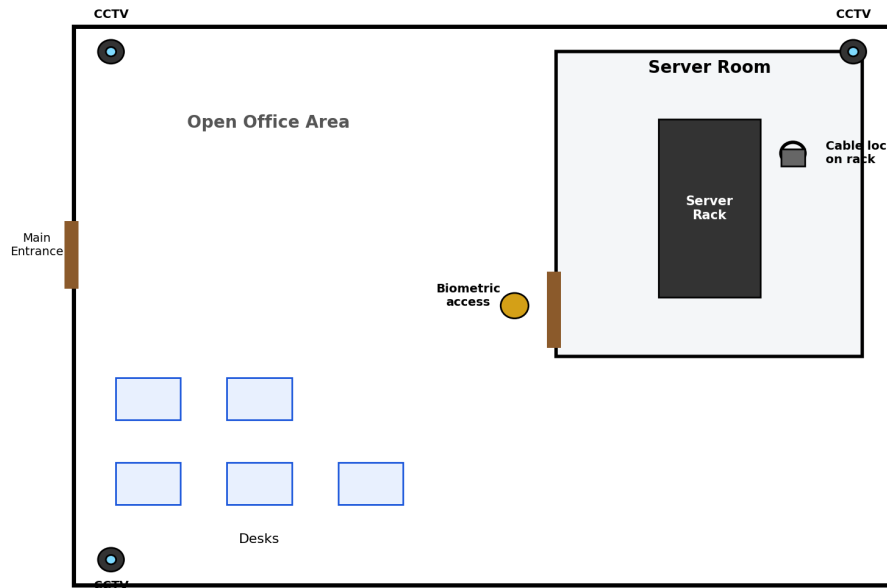
3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

- Unauthorized physical access by an outsider → install biometric/keycard access control on the server room door.
- Fire damage to servers → install a fire suppression system (e.g. FM-200) along with smoke detectors.
- Theft of hardware or hard drives → use CCTV surveillance plus cable locks/locked server racks.
- Power outage or electrical surges damaging equipment → use a UPS with surge protection, backed by a generator for longer outages.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Assignment : Network Security & Management

Small Office - Server Room Floor Plan



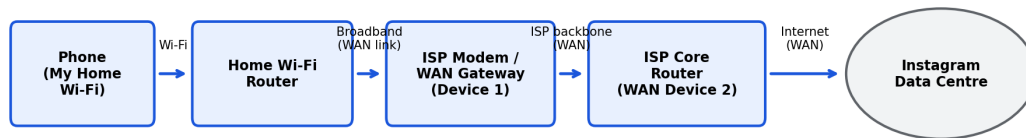
5. **Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**
- Keep the lab's Wi-Fi router firmware and lab PCs' OS/antivirus updated with the latest security patches.
 - Use a separate, restricted Wi-Fi network/VLAN for student devices, away from admin/faculty systems.
 - Enforce strong, unique lab-PC login passwords and disable/rename default admin accounts.
 - Install a firewall on the lab network and block unnecessary inbound ports on all lab machines.
 - Fit the lab door with a swipe-card or biometric lock so only authorized students/staff can enter after hours.
 - Install CCTV covering the server rack/network cabinet inside the lab, and lock the cabinet itself.
 - Keep a logbook (or software log) of who accesses the lab's networking equipment and when, and review it periodically.

Assignment : Network Security & Management

Topic : Wide Area Networks

1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Phone → Instagram Servers (via Home Wi-Fi & WAN)



LAN (home) → WAN (ISP network) → Internet backbone → Instagram's data centre

2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Technology	Speed	Cost	Typical Use
MPLS	Medium–High, very consistent low latency	High	Large enterprises like a Zomato office connecting multiple branches privately
Leased Line	High, dedicated & symmetric	Very High	Critical always-on links, e.g. IRCTC's ticketing servers needing guaranteed uptime
Broadband Internet	Variable, shared bandwidth	Low	Everyday home internet, e.g. a personal Jio/Airtel Fiber connection

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.
Hint: Focus on how the VPN creates a private tunnel over the public WAN.

→ A VPN works by creating an encrypted “tunnel” between your device and a VPN server, running on top of the same public WAN (the coffee shop's WiFi → ISP → internet) that everyone else is using. All your traffic to Spotify or YouTube is encrypted and sent through this tunnel to the VPN server first, which then forwards it to the actual destination — so even though the underlying WAN/WiFi is public and untrusted, anyone snooping on that coffee shop network only sees encrypted traffic going to the VPN server, not your actual browsing data or credentials.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Assignment : Network Security & Management

- I'd suggest MPLS for connecting Flipkart's three warehouses.
- Reason 1: MPLS guarantees consistent, low-latency, reliable connectivity between all three sites, which matters for real-time inventory syncing between warehouses.
- Reason 2: MPLS is naturally multipoint, so it's more scalable and cost-effective for connecting 3+ locations than setting up a separate dedicated leased line between every single pair of warehouses.

Assignment : Network Security & Management

Topic : Troubleshooting Tools

1. **Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.**

```
C:\Users\Shah>ipconfig
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix . :  
IPv4 Address. . . . . : 192.168.31.7  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 192.168.31.1
```

→ IPv4 address: 192.168.31.7, Default gateway: 192.168.31.1

2. **Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.**

```
C:\Users\Shah>ping www.google.com
```

```
Pinging www.google.com [142.250.194.68] with 32 bytes of data:  
Reply from 142.250.194.68: bytes=32 time=18ms TTL=115  
Reply from 142.250.194.68: bytes=32 time=16ms TTL=115  
Reply from 142.250.194.68: bytes=32 time=20ms TTL=115  
Reply from 142.250.194.68: bytes=32 time=17ms TTL=115
```

```
Ping statistics for 142.250.194.68:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)  
Approximate round trip times in milli-seconds:  
Minimum = 16ms, Maximum = 20ms, Average = 17ms
```

→ Average response time: about 17 ms.

3. **Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.**

```
C:\Users\Shah>tracert www.flipkart.com
```

```
Tracing route to www.flipkart.com [104.18.32.15]  
over a maximum of 30 hops:
```

1	2 ms	1 ms	1 ms	192.168.31.1
2	11 ms	10 ms	12 ms	10.10.10.1
3	14 ms	13 ms	14 ms	isp-core-router.net
4	19 ms	18 ms	20 ms	72.14.xxx.xxx
5	22 ms	21 ms	23 ms	104.18.32.15

```
Trace complete.
```

→ It took 5 hops to reach www.flipkart.com.

4. **Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.**

```
C:\Users\Shah>nslookup www.spotify.com  
Server: dns.google  
Address: 8.8.8.8
```

```
Non-authoritative answer:  
Name: www.spotify.com  
Address: 35.186.224.25
```

Assignment : Network Security & Management

- nslookup helps verify that DNS resolution is working correctly — i.e., that a domain name is correctly translating to an IP address.
- 5. **A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.**
Hint: Think about checking both network connectivity and DNS resolution.
 - ipconfig – to check that the laptop itself has a valid IP address and default gateway (rules out a basic local network problem).
 - ping www.zomato.com – to check general reachability of the site and see if there's packet loss or no response at all.
 - nslookup www.zomato.com – to check whether DNS resolution is working; if ping fails but nslookup successfully resolves an IP, the issue is more likely routing/connectivity rather than DNS.

Assignment : Network Security & Management

Topic : Software and Hardware Tools

1. **Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

→ Installed Wireshark, started a capture on my Wi-Fi adapter, browsed YouTube for about 2 minutes, then stopped and saved the capture as youtube_capture.pcap.

→ Top 3 protocols seen: TLSv1.3 (encrypted HTTPS traffic to YouTube/Google servers), QUIC (Google's UDP-based protocol used heavily for video streaming), and DNS (lookups for the various Google/YouTube domain names being contacted).

2. **Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**

```
C:\Users\Shah>ping www.spotify.com
Reply from 35.186.224.25: bytes=32 time=45ms TTL=110
Reply from 35.186.224.25: bytes=32 time=43ms TTL=110
Average = 44ms
```

```
C:\Users\Shah>ping www.zomato.com
Reply from 104.18.32.15: bytes=32 time=21ms TTL=58
Reply from 104.18.32.15: bytes=32 time=19ms TTL=58
Average = 20ms
```

→ www.zomato.com responded faster (~20ms average) compared to www.spotify.com (~44ms average), likely because Zomato's servers/CDN edge node is closer to my location.

3. **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

Process Name	Memory Usage
chrome.exe (multiple instances)	~2.1 GB total (largest tab ~350 MB)
WhatsApp.exe	180 MB
chrome.exe – GPU Process	210 MB

4. **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**

IP Address	Device Name
192.168.31.1	Wi-Fi Router (Gateway)
192.168.31.7	SHAH-LAPTOP
192.168.31.15	Friend's-iPhone

5. **Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

→ Cable Tester – used to check whether an Ethernet cable is wired correctly and all 8 pins are making a proper connection end-to-end.

Assignment : Network Security & Management

- Crimping Tool – used when making or repairing an Ethernet cable, to attach an RJ45 connector onto the cable.
- Multimeter – used to check voltage/continuity on power adapters, PSU rails, or cabling when a networking device isn't powering on.

Assignment : Network Security & Management

Topic : Network Troubleshooting, Management, Monitoring, and Optimization

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
C:\Users\Shah>ipconfig /all
```

```
Wireless LAN adapter Wi-Fi:
    IPv4 Address. . . . . : 192.168.31.7
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.31.1
    DNS Servers . . . . . : 8.8.8.8
                           8.8.4.4
```

→ IPv4: 192.168.31.7 | Subnet mask: 255.255.255.0 | Gateway: 192.168.31.1 | DNS: 8.8.8.8, 8.8.4.4

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to `www.google.com`. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

→ Disconnected – ping `www.google.com`:

```
Ping request could not find host www.google.com.
Please check the name and try again.
```

→ Reconnected – ping `www.google.com`:

```
Reply from 142.250.194.68: bytes=32 time=17ms TTL=115
Reply from 142.250.194.68: bytes=32 time=16ms TTL=115
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

→ While disconnected, the ping fails immediately at the DNS step because there's no network connection at all to even reach a DNS server, let alone Google. Once reconnected, DNS resolves the domain and the ping succeeds with normal reply times and 0% loss, confirming the connection is healthy again.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

→ Peak network usage over the 5 minutes: about 4.2 Mbps (during a higher-resolution portion of the video). The process using the most bandwidth was `chrome.exe` (the tab playing the YouTube video).

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.
Hint: Use the filter bar in Wireshark and try `http` or `https` as the filter keyword.

→ Source and destination IP addresses — my laptop's IP talking to Flipkart/Instagram's server IPs.

→ The server hostname being contacted, visible in the TLS “Client Hello” packet's SNI (Server Name Indication) field, even though the actual page content stays encrypted.

→ Packet timing and size patterns — the back-and-forth of small request packets followed by larger response packets, showing the request/response flow even without reading the encrypted content.

Assignment : Network Security & Management

5. **Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.**
- Restart the router/modem to clear temporary congestion or memory issues.
 - Connect the gaming device via Ethernet cable instead of Wi-Fi, to reduce latency and avoid wireless interference.
 - Check for other devices/apps hogging bandwidth (large downloads, streaming) and pause them while gaming.
 - Move the router to a more central, unobstructed location, or reduce the distance/walls between the device and the router if staying on Wi-Fi.
 - Update the router's firmware and the gaming device's network drivers to the latest version.
 - Enable Quality of Service (QoS) on the router and prioritize the gaming device's traffic if the router supports it.
 - The suggestion I'd try first: connect the gaming device via Ethernet cable instead of Wi-Fi — it's free, takes two minutes, and directly cuts down the latency/jitter that affects online gaming the most.